Yee-Cell Finite-Difference Solver for 2D Electromagnetics

Formulation, Discretization, and Implementation Plan
with a Roadmap to GSTC Metasurfaces

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Abstract

This note is the specification for MaxwellFD, a Python finite-difference code on a uniform Cartesian Yee grid. One grid, one material sampling, and one pair of discrete curls serve both the leapfrog scheme (FDTD) and the time-harmonic sparse scheme (FDFD). Phasors use $e^{j\omega t}$, the same convention as the sibling surface code MaxwellMOM. The derivation below is two-dimensional, for TMz and TEz, in free space and in a uniform isotropic medium. PEC and periodic walls, the CFL limit, numerical dispersion, a stretched-coordinate PML, and a resistive sheet are written out against equation numbers. The resistive conductance is in the code. The non-dispersive GSTC subcell is the symmetric Yee cell: the face unknowns and the local system are derived here, and the tangential sheet is in the code. A Bloch phase on the periodic x wrap is in the code. A piecewise sheet assigns each cut face its own susceptibilities. Polynomial electric dispersion is in the code as the normal susceptibility $\chi^{nn}_{mm} = \chi_{ee2}k^2$. Polynomial magnetic dispersion is the second difference of tangential H on the electric jump. A denominator in k_t remains a check.

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1 Mathematical formulation

1.1 Maxwell's equations

Fields are in SI units. The time convention is $e^{j\omega t}$, so a wave $e^{-jk\hat{\mathbf{k}}\cdot\mathbf{r}}$ propagates in the direction $\hat{\mathbf{k}}$ and ∂_t becomes $j\omega$. With an impressed current $\mathbf{J}$,

$$\nabla \times \mathbf{E} = -j\omega\mu\,\mathbf{H}, \tag{1.1}$$

$$\nabla \times \mathbf{H} = j\omega\varepsilon\,\mathbf{E} + \mathbf{J}. \tag{1.2}$$

Conduction is not a second current on the right. It stays in a real conductivity σ that the time stepper absorbs into its coefficients, or, in a later frequency-domain material, in a complex permittivity. The constants c_0, ε_0, μ_0, and η_0 are the SI values used by MaxwellMOM. In a uniform lossless medium $c = 1/\sqrt{\mu\varepsilon}$ and $\eta = \sqrt{\mu/\varepsilon}$.

The impressed current is the source. A z-directed Hertzian dipole places the moment $I\ell$ on one E_z sample, $J_z = I\ell/(\Delta x\Delta y\Delta z)$, and the 3D solve compares the electric field with the outgoing spherical wave. A 2D paraxial Gaussian beam places its impressed current on the two E_z rows that touch a horizontal split. Below the split the unknown is the scattered field, and the total field above the split is compared with the beam. A 2D TMz near-to-far integral reads E_z and the tangential H on a closed contour in the unstretched region. With $\mathbf{n}$ the outward normal and $K_z = (\mathbf{n}\times\mathbf{H})_z$, the far-zone field is the leading $H_0^{(2)}$ factor times the integral of $e^{jk\hat{\boldsymbol{\rho}}\cdot\mathbf{r}'}\big[j\omega\mu K_z - jk(\mathbf{n}\cdot\hat{\boldsymbol{\rho}})E_z\big]$. A line current at the origin returns $-(\omega\mu I/4)$ times that factor. On a Yee rectangle, E_z is the node value and the tangential H is the average of the two faces that touch the edge. Milestone 1 does not inject those sources. Its free-space check is a periodic eigenmode, and its cavity check is a source-free resonance.

1.2 Two-dimensional reductions

The grid lies in the xy-plane and nothing depends on z.

TMz. The unknowns are E_z, H_x, and H_y. Equations (1.1) and (1.2) reduce to

$$\partial_y E_z + j\omega\mu H_x = 0, \tag{1.3}$$

$$-\partial_x E_z + j\omega\mu H_y = 0, \tag{1.4}$$

$$\partial_x H_y - \partial_y H_x - j\omega\varepsilon E_z = J_z. \tag{1.5}$$

Eliminating $\mathbf{H}$ gives the Helmholtz equation used only as a check,

$$\partial_x^2 E_z + \partial_y^2 E_z + \omega^2\mu\varepsilon E_z = j\omega\mu J_z. \tag{1.6}$$

The code never forms this scalar operator when μ varies from face to face. It keeps the curls.

TEz. The unknowns are H_z, E_x, and E_y:

$$\partial_x E_y - \partial_y E_x + j\omega\mu H_z = 0, \tag{1.7}$$

$$\partial_y H_z - j\omega\varepsilon E_x = J_x, \tag{1.8}$$

$$-\partial_x H_z - j\omega\varepsilon E_y = J_y. \tag{1.9}$$

1.3 Constitutive law

A volumetric law returns ε, μ, and σ at the Yee location of each component. The milestone-1 law is uniform and isotropic: one positive ε, one positive μ, and one $\sigma \geq 0$. The call also accepts a frequency and an external state (a temperature map, a stress map, or anything the caller has

already computed). The uniform law ignores both. A later law may return a different value at every sample, or a complex $\varepsilon(\omega)$ for FDFD. The time stepper requires real ε and σ; a dispersive time-domain medium will add auxiliary variables rather than a complex coefficient inside the leapfrog.

Surface models are not constitutive samples. PEC removes unknowns. A resistive sheet adds the conductance of Sec. 4.3 to one Ampere cell. A GSTC sheet replaces the magnetic sample on the cut. None of these is encoded by setting ε to zero or to infinity.

1.4 Boundaries in the continuous problem

On a PEC wall the tangential electric field is zero. On a PMC wall the tangential magnetic field is zero. A periodic wall identifies the two opposite faces. The coded phase on a forward x wrap is $e^{-jk_{B,x}a}$, and the backward wrap carries $e^{+jk_{B,x}a}$. Milestone 1 leaves $k_{B,x} = 0$, so that factor is one. The y wrap stays at phase one. An open boundary is a PML, the stretch of Sec. 3.5.

2 Discretization

2.1 Cells, axes, and shapes

The grid has n_x by n_y cells of size Δx and Δy. The domain is $[0, a] \times [0, b]$ with $a = n_x \Delta x$ and $b = n_y \Delta y$. Stored arrays use index $[i, j]$ with axis 0 along x.

Table 1 is the layout the drivers assume. PEC keeps the boundary electric samples in the array and sets them to zero. Periodic identification drops the duplicate far side, so every stored sample is an unknown and every difference wraps.

Table 1: Sample locations. A PEC count of n+1 includes both boundaries. A periodic count of n is one period.

Field	Location	PEC shape	Periodic shape
E_z	$(i\Delta x,\ j\Delta y)$	$(n_x{+}1, n_y{+}1)$	(n_x, n_y)
H_x	$(i\Delta x,\ (j+\frac{1}{2})\Delta y)$	$(n_x{+}1, n_y)$	(n_x, n_y)
H_y	$((i+\frac{1}{2})\Delta x,\ j\Delta y)$	$(n_x, n_y{+}1)$	(n_x, n_y)
H_z	$((i+\frac{1}{2})\Delta x,\ (j+\frac{1}{2})\Delta y)$	(n_x, n_y)	(n_x, n_y)
E_x	$((i+\frac{1}{2})\Delta x,\ j\Delta y)$	$(n_x, n_y{+}1)$	(n_x, n_y)
E_y	$(i\Delta x,\ (j+\frac{1}{2})\Delta y)$	$(n_x{+}1, n_y)$	(n_x, n_y)

Centered differences of a linear field are exact on this stencil. That is the sign test for the curls: if $E_z = \alpha x + \beta y$, then the discrete $\partial_y E_z$ equals β and $-\partial_x E_z$ equals $-\alpha$.

2.2 Discrete curls

Collect the free electric samples in a vector e and the magnetic samples that the curls touch in a vector h. The two real sparse matrices are defined so that they match (1.1) and (1.2) term by term:

$$\mathrm{curl}_E\, \mathrm{e} + j\omega\, \mathrm{diag}(\mu_H)\, \mathrm{h} = 0, \tag{2.1}$$

$$\mathrm{curl}_H\, \mathrm{h} - j\omega\, \mathrm{diag}(\varepsilon_E)\, \mathrm{e} = \mathrm{j}. \tag{2.2}$$

For TMz, curl_E stacks $(\partial_y E_z,\ -\partial_x E_z)$ and curl_H is $\partial_x H_y - \partial_y H_x$. For TEz, curl_E is $\partial_x E_y - \partial_y E_x$ and curl_H stacks $(\partial_y H_z,\ -\partial_x H_z)$.

On every grid in Table 1 the two matrices are transposes,

$$\mathrm{curl}_H = \mathrm{curl}_E^{\mathsf{T}}. \tag{2.3}$$

The boundary terms that integration by parts would leave are absent because PEC electric samples are not unknowns, and periodic faces cancel. The product $\mathrm{curl}_H \,\mathrm{diag}(\mu^{-1})\,\mathrm{curl}_E$ is therefore symmetric. A nonzero $k_{B,x}$ makes both curls complex and replaces that transpose by the conjugate transpose, so the product is Hermitian. The y wrap stays at phase one. Every grid in Table 1 keeps $k_{B,x} = 0$, and (2.3) is that case.

PEC degrees of freedom, TMz: E_z for $i = 1, \ldots, n_x - 1$ and $j = 1, \ldots, n_y - 1$; H_x for $i = 1, \ldots, n_x - 1$ and $j = 0, \ldots, n_y - 1$; H_y for $i = 0, \ldots, n_x - 1$ and $j = 1, \ldots, n_y - 1$. The magnetic samples on the outer PEC edges do not enter curl_H at a free E_z. TEz keeps every H_z, drops E_x on $y = 0$ and $y = b$, and drops E_y on $x = 0$ and $x = a$.

TEz curl-curl on tangential $\mathbf{E}$ has a kernel: any discrete gradient of a nodal potential that is constant on the PEC boundary has zero circulation. Those eigenvalues are zero. They are not cavity modes. The physical spectrum is the positive one.

The 3D cell uses the same partial derivatives on every axis. Sample locations extend Table 1: E_x sits at $((i + \frac{1}{2})\Delta x,\, j\Delta y,\, k\Delta z)$, and each electric component is staggered along its own direction. The magnetic components are staggered along the other two. A PEC wall drops the tangential electric samples. Magnetic samples that do not touch a free electric sample are omitted, so (2.3) still holds on a PEC box and on a periodic box.

2.3 Leapfrog update

Store $\mathbf{E}$ at integer steps and $\mathbf{H}$ at the half-step before it. One step produces $\mathbf{H}^{n+1/2}$ from $\mathbf{E}^n$, then $\mathbf{E}^{n+1}$ from $\mathbf{H}^{n+1/2}$. For TMz, with the curls of Sec. 2.2,

$$H_x \leftarrow H_x - \frac{\Delta t}{\mu_x}\,\partial_y E_z, \tag{2.4}$$

$$H_y \leftarrow H_y + \frac{\Delta t}{\mu_y}\,\partial_x E_z, \tag{2.5}$$

$$E_z \leftarrow C_a E_z + C_b\,(\partial_x H_y - \partial_y H_x). \tag{2.6}$$

The coefficients are evaluated at the E_z samples,

$$C_a = \frac{1 - \sigma\Delta t/(2\varepsilon)}{1 + \sigma\Delta t/(2\varepsilon)}, \qquad C_b = \frac{\Delta t/\varepsilon}{1 + \sigma\Delta t/(2\varepsilon)}. \tag{2.7}$$

When $\sigma = 0$ they are $C_a = 1$ and $C_b = \Delta t/\varepsilon$. The same pair is used on E_x and E_y in TEz, and the magnetic update is

$$H_z \leftarrow H_z - \frac{\Delta t}{\mu_z}(\partial_x E_y - \partial_y E_x). \tag{2.8}$$

After the electric update, PEC edges are set back to zero.

No matrix is stored. The array curls are the same stencil as curl_E and curl_H, including the wrap on a periodic grid.

2.4 Material sampling and staircase geometry

ε and σ live on electric samples. μ lives on magnetic samples. A uniform law fills those arrays with constants. A staircase solid fills each sample from the coordinate of that sample, so an electric node and a magnetic node on the same cell can lie on opposite sides of a face. A slab $y_0 \le y < y_1$ is half-open, and a disk $(x - x_c)^2 + (y - y_c)^2 \le R^2$ is closed. Later regions overwrite earlier ones. There is no volume-fraction weight. The assignment is

$$(x, y) \in \mathcal{R} \quad \Longrightarrow \quad (\varepsilon, \mu, \sigma)(x, y) = (\varepsilon, \mu, \sigma)_{\mathcal{R}}, \tag{2.9}$$

The stepper is unchanged, because it already reads one coefficient per sample. Conformal cuts and sub-cell dielectric smoothing are not part of this scheme.

Tensor $\bar{\varepsilon}$ is out of scope until the scalar isotropic tests match the analytic formulas. Off-diagonal samples would sit at locations the present curls do not read.

2.5 Bulk Debye and Lorentz media

A dispersive law returns a complex $\varepsilon(\omega)$ to FDFD and a real instantaneous permittivity ε_∞ to the leapfrog. One electric Debye pole and one electric Lorentz pole are enough for this slice. With the $e^{j\omega t}$ convention,

$$\varepsilon(\omega) = \varepsilon_\infty + \frac{\varepsilon_0 \Delta\varepsilon}{1 + j\omega\tau}, \tag{2.10}$$

$$\tau \partial_t P + P = \varepsilon_0 \Delta\varepsilon \, E, \tag{2.11}$$

$$\varepsilon(\omega) = \varepsilon_\infty + \frac{\varepsilon_0 \omega_p^2}{\omega_0^2 - \omega^2 + j\gamma\omega}, \tag{2.12}$$

$$\partial_t^2 P + \gamma \partial_t P + \omega_0^2 P = \varepsilon_0 \omega_p^2 E. \tag{2.13}$$

Magnetic dispersion is not in this slice. These volume auxiliaries are independent of the surface oscillators (6.17) and (6.18).

$J_p = \partial_t P$ is added to the impressed current. The coefficients C_a and C_b stay (2.7), built on ε_∞ and σ, and the electric update becomes

$$E \leftarrow C_a E + C_b \, (\nabla \times H - J_p - J). \tag{2.14}$$

$J = 0$ and $J_p = 0$ is (2.6). Lorentz stores P at integer steps, advances it explicitly from E^n, and sets $J_p^{n+1/2} = (P^{n+1} - P^n)/\Delta t$. Debye uses the centered average $E^{n+1/2} = (E^{n+1} + E^n)/2$, which couples P^{n+1} to E^{n+1} and is solved one sample at a time. TEz carries one auxiliary on E_x and one on E_y.

3 Stability, dispersion, and domain closure

3.1 CFL limit

On a uniform 2D Yee grid the leapfrog is stable for

$$\Delta t \le \Delta t_{\mathrm{CFL}} = \frac{1}{c\sqrt{\Delta x^{-2} + \Delta y^{-2}}}. \tag{3.1}$$

The bound is sharp for the grid mode $k_x = \pi/\Delta x$, $k_y = \pi/\Delta y$. A run at $0.99\,\Delta t_{\mathrm{CFL}}$ must stay bounded. A run at $1.01\,\Delta t_{\mathrm{CFL}}$ must grow.

3.2 Semi-discrete and fully discrete frequencies

With continuous time and Yee differences, a mode k_x, k_y oscillates at

$$\left(\frac{\omega_0}{c}\right)^2 = \left(\frac{2\sin(k_x \Delta x/2)}{\Delta x}\right)^2 + \left(\frac{2\sin(k_y \Delta y/2)}{\Delta y}\right)^2. \tag{3.2}$$

The leapfrog moves that frequency to the root of

$$\left(\frac{\sin(\omega \Delta t/2)}{c\,\Delta t}\right)^2 = \left(\frac{\sin(k_x \Delta x/2)}{\Delta x}\right)^2 + \left(\frac{\sin(k_y \Delta y/2)}{\Delta y}\right)^2, \tag{3.3}$$

provided the right-hand side is at most $(c\Delta t)^{-2}$. Because $\arcsin z > z$ on $(0, 1)$, this ω lies slightly above ω_0. FDFD, having no time difference, must reproduce ω_0.

A rectangular PEC cavity of size $a \times b$ has $k_x = m\pi/a$ and $k_y = n\pi/b$. TMz needs $m \ge 1$ and $n \ge 1$, because E_z vanishes on the whole boundary. TEz also allows $m = 0$ or $n = 0$, but not both. The mode shapes consistent with the stencil are

$$E_z = \sin(m\pi x/a)\sin(n\pi y/b), \tag{3.4}$$

$$H_z = \cos(m\pi x/a)\cos(n\pi y/b). \tag{3.5}$$

A rectangular box $a \times b \times c$ adds the term $(2\sin(k_z \Delta z/2)/\Delta z)^2$ to (3.2), with $k_z = p\pi/c$. The indices are non-negative and at least two of them are positive. A mode with exactly one index equal to zero is a single Cartesian component: $(m, n, 0)$ is $E_z = \sin(m\pi x/a)\sin(n\pi y/b)$, $(m, 0, p)$ is E_y, and $(0, n, p)$ is E_x. These three are exact eigenmodes of the centered difference. A mode with three positive indices has two polarizations linked by $k_x A_x + k_y A_y + k_z A_z = 0$.

A pure spatial eigenmode sampled on three successive steps obeys the scalar leapfrog recurrence. Those three numbers determine ω without a Fourier transform.

3.3 Lossless invariant

E and **H** are never stored at the same time. The sum of their sample energies therefore oscillates by a noticeable fraction of itself near the CFL limit, even though the scheme is lossless. The quadratic form that the update conserves, when $\sigma = 0$, is

$$W^n = \frac{1}{2}\sum \varepsilon\,(E^n)^2\,\Delta x \Delta y + \frac{1}{2}\sum \mu\, H^{n+1/2} H^{n-1/2}\,\Delta x \Delta y, \tag{3.6}$$

with the sums over the stored samples of each component. The derivation is the adjoint relation (2.3): the work done on **E** by $\mathrm{curl}_H \mathbf{H}$ cancels the work done on **H** by $\mathrm{curl}_E \mathbf{E}$. Conductivity removes the cancellation. Above the CFL limit W^n can stay constant while the ordinary field energy grows, because W^n is then indefinite. Boundedness is checked with W^n. Growth past the CFL limit is checked with the ordinary field energy.

3.4 PEC, PMC, and periodic walls

PEC for TMz is $E_z = 0$ on the outer node lines. PEC for TEz is $E_x = 0$ on $y \in \{0, b\}$ and $E_y = 0$ on $x \in \{0, a\}$. PMC is the dual statement, tangential $\mathbf{H} = 0$, which is a Neumann condition on the remaining field. PMC is part of the surface stack and is not in the milestone-1 unknown set.

Periodic differences replace $i - 1$ at $i = 0$ by $i = n_x - 1$, and likewise in y. The forward x image, the neighbor past $i = n_x - 1$, is multiplied by $e^{-jk_{B,x}a}$. The backward image, the neighbor before $i = 0$, is multiplied by $e^{+jk_{B,x}a}$. The phase is one when $k_{B,x} = 0$. The y wrap is unchanged. The leapfrog and the real eigenproblem refuse a nonzero $k_{B,x}$.

3.5 PML

An open domain is closed by a complex stretch of the normal coordinate, following the convolutional PML. In the frequency domain, for each axis $w \in \{x, y\}$,

$$s_w = \kappa_w + \frac{\sigma_w}{\alpha_w + j\omega\varepsilon_0}, \tag{3.7}$$

and every ∂_w in the curls is replaced by $s_w^{-1}\partial_w$. The reciprocal that the auxiliary equation integrates is

$$\frac{1}{s_w} = \frac{1}{\kappa_w} - \frac{\sigma_w/(\kappa_w^2\varepsilon_0)}{j\omega + \alpha_w/\varepsilon_0 + \sigma_w/(\kappa_w\varepsilon_0)}. \tag{3.8}$$

Inside a PML slab of thickness d, with ρ the depth measured from the inner face and m typically 3 or 4,

$$\sigma_w(\rho) = \sigma_{\max}(\rho/d)^m, \qquad \sigma_{\max} = -\frac{(m+1)\ln R(0)}{2\eta_0 d}, \tag{3.9}$$

with a target reflection $R(0)$ of order 10^{-6} to 10^{-8}. κ_w grows from 1 at the inner face to a modest $\kappa_{\max}$, and α_w falls from a small positive value at the inner face to zero at the outer PEC backing. The time-domain scheme adds one auxiliary per stretched derivative and integrates (3.8) with the same Δt as the leapfrog. `operators.pml` is that stretch, on a PEC-backed band, in both drivers. A solve with no PML object is the milestone-1 stepper. The regression compares a line current with the outgoing $H_0^{(2)}$ field on the unstretched samples.

4 Linear algebra and thin sheets

4.1 Driven FDFD system

Eliminating h from (2.1) and (2.2) gives the system that is factored,

$$\operatorname{curl}_H \operatorname{diag}\left(\frac{1}{j\omega\mu_H}\right)\operatorname{curl}_E \mathrm{e} + j\omega \operatorname{diag}(\varepsilon_E)\,\mathrm{e} = -\mathrm{j}. \tag{4.1}$$

The unknown is the free electric vector. PEC rows are omitted rather than penalized. The matrix is complex symmetric when ε and μ are scalar, and it is stored sparse. Milestone 1 solves it with a sparse direct factorization. Two-dimensional grids of the size used in the tests factor in negligible time. Iterative solves (GMRES or BiCGSTAB, with a preconditioner) wait until a grid is large enough that the factor is the expensive part.

At a fixed real ω away from the cavity spectrum the matrix is nonsingular. A manufactured solution builds j by applying the left-hand side to a chosen e and asks the solver for that e back.

4.2 Cavity eigenproblem

For lossless media and $\mathrm{j} = 0$, multiplying (4.1) through by $j\omega$ produces the generalized eigenproblem

$$K\mathrm{e} = \omega^2 M\mathrm{e}, \qquad K = \operatorname{curl}_H \operatorname{diag}(\mu_H^{-1})\operatorname{curl}_E, \qquad M = \operatorname{diag}(\varepsilon_E). \tag{4.2}$$

K is real symmetric and, on a TMz PEC cavity, positive definite. On a TEz PEC cavity it is positive semi-definite with the gradient kernel of Sec. 2.2. The positive eigenvalues are the semi-discrete frequencies (3.2). Conductivity does not enter K and M. A complex permittivity belongs in the driven system (4.1), not in this real eigenproblem.

4.3 Resistive sheet

A zero-thickness electric sheet with surface impedance Z_s (ohms per square) carries

$$\hat{\mathbf{n}} \times (\mathbf{H}^+ - \mathbf{H}^-) = \mathbf{E}_{\mathrm{av}}/Z_s. \tag{4.3}$$

The normal points from the $-$ side to the $+$ side. For a sheet of normal $\hat{\mathbf{y}}$ in TMz this is $-(H_x^+ - H_x^-) = E_z/Z_s$.

The same sheet is the pure-electric GSTC of Sec. 6 with

$$\chi_{ee} = \frac{\eta_0}{jkZ_s} = \frac{1}{j\omega\varepsilon_0 Z_s}, \tag{4.4}$$

and $\chi_{mm} = 0$. MaxwellMOM's `chi_ee_from_Zs` is this map. In the time domain a real Z_s is a shunt conductance, not a constant susceptibility: $J_z = \sigma_s E_z$ with $\sigma_s = 1/Z_s$. Placing that current in the Ampere cell of height Δy that contains the sheet adds

$$\sigma_{\mathrm{eff}} = \sigma + \frac{1}{Z_s \Delta y} \tag{4.5}$$

to the C_a, C_b formula (2.7) on the sheet's electric samples. That is the milestone-4 update. It is the $\chi_{mm} = 0$ reduction of the symmetric cell, specialized to $\chi_{ee} \propto 1/(j\omega)$.

4.4 Complex permittivity and a Dirichlet trace

When ε and σ are real, the diagonal of (4.1) uses

$$\varepsilon_{\mathrm{FDFD}} = \varepsilon + \frac{\sigma}{j\omega}. \tag{4.6}$$

A law that returns a complex $\varepsilon(\omega)$ has already included its loss. That array replaces (4.6) and the accompanying σ is zero, so a pole is not counted twice. The real eigenproblem (4.2) still requires real positive ε and μ and $\sigma = 0$.

PEC electric samples stay out of the unknown vector. A non-zero trace e_D, equal to the prescribed boundary values and zero on the free samples, moves onto the right-hand side,

$$A_{ff}\, \mathrm{e}_f = -\mathrm{j} - (A\mathrm{e}_D)_f. \tag{4.7}$$

The free rows of $A\mathrm{e}_D$ are the array form of the left-hand side of (4.1). A zero trace is the milestone-1 solve. This is how the slab and cylinder comparisons are posed. It is not a PML.

An embedded PEC uses the same omission on every electric sample whose coordinate lies inside the object. An embedded PMC omits the magnetic samples in its region, so the tangential H on a staircase face is zero and the magnetic samples of a solid interior leave the unknown vector. Both are masks on the Yee locations. Setting ε to infinity, or μ to zero, is a different model. The open PEC cylinder puts $e_D = -E_{\mathrm{inc}}$ on the object and $e_D = 0$ on the outer wall, and the PML absorbs the scattered field. Inside the object the total electric field is zero.

An open dielectric cylinder keeps that PML and the homogeneous outer wall. It does not remove interior samples. The staircase permittivity stays on the diagonal of (4.1), and the incident wave enters as the impressed current $J = j\omega(\varepsilon - \varepsilon_0)E_{\mathrm{inc}}$ on the electric samples. The current is zero wherever $\varepsilon = \varepsilon_0$. A free sample of the total field is $E_s + E_{\mathrm{inc}}$. Inside the disk the total field is the Bessel series of Sec. 5.2.

5 Validation

Milestone 1 is a numerical check that the curls, the leapfrog, and the sparse system implement Secs. 2 and 3. The acceptance tests and the bars they use are:

Check	Bar	What is compared
Linear TMz and TEz curls	10^{-12} absolute	Centered differences of a linear field
$\mathrm{curl}_H = \mathrm{curl}_E^{\mathsf{T}}$	10^{-12} absolute	All four grids in Table 1
Matrix versus array curl	10^{-12} relative	Same four grids, PEC boundaries held at zero
TMz cavity $(1, 1)$	10^{-10} relative	Smallest FDFD eigenvalue versus (3.2)
TEz cavities $(1, 0)$ and $(1, 1)$	10^{-10} relative	Positive FDFD eigenvalues versus (3.2)
Manufactured FDFD solve	10^{-10} relative	Recovered e on TMz and TEz
Leapfrog invariant at $0.99\,\Delta t_{\mathrm{CFL}}$	1 percent	W^n over $2 \cdot 10^3$ steps, random TMz data
Field energy at $1.01\,\Delta t_{\mathrm{CFL}}$	growth past $10\times$	Same data, ordinary sample energy
TMz and TEz modal ω	10^{-6} relative	Three leapfrog samples versus (3.3)
Periodic standing wave $(k_x, k_y) = (2\pi/a,\, 0)$	10^{-6} relative	Same recurrence versus (3.3)

The modal and standing-wave bars in the project plan were 10^{-3}, which is a fit tolerance. A pure eigenmode has no fit error, so the tests use 10^{-6}.

Milestone 2 adds the volumetric checks. The scattering rows compare the Yee solution, with the trace (4.7) taken from the formulas below, to those formulas. The gap is dispersion, and for the disk the staircase. A slab face used in a convergence pair is kept off the electric nodes of both grids, so (2.9) does not assign the face to a sample. The comparison wavelength is kept off the TEz eigenfrequencies of that closed box: $\lambda = 40\Delta$ on the 8×80 domain is one of them, and the Dirichlet problem is then singular.

Check	Bar	What is compared
Fresnel face continuity	10^{-12} absolute	Tangential E and H, both polarizations
Mie cylinder continuity	10^{-8} absolute	Tangential field and derivative at $\rho = a$
PEC cylinder coefficient	10^{-12} relative	a_n against $-(-j)^n J_n / H_n^{(2)}$
Manufactured $\sigma > 0$ and $\varepsilon(\mathbf{r})$	10^{-10} relative	Recovered e, both polarizations
Uniform $\varepsilon_r = 4$ cavity	10^{-10} / 10^{-6}	FDFD versus (3.2); leapfrog versus (3.3)
Slab-loaded TMz cavity	10^{-2} and 10^{-6}	Continuous root, then leapfrog against that ω_0
Lossless staircase invariant	1 percent	W^n over 500 steps at half the CFL step
Fresnel slab, both polarizations	$2 \cdot 10^{-2}$ relative	Free samples; halving Δ drops the error by 2.5×
Dielectric cylinder, both polarizations	$2 \cdot 10^{-2}$ relative	Samples 2Δ off the circle and the wall
Debye and Lorentz, one period	$3 \cdot 10^{-3}$ relative	Leapfrog against the complex-ε phasor

The convolutional PML is checked apart from those Dirichlet rows. The profile row is the grading of Sec. 3.5. The line-current rows compare E_z of a unit line current with the outgoing $H_0^{(2)}$ field on the unstretched samples at least $\lambda/2$ from the source. Halving the spacing keeps the same physical band. The bare-box row uses that same window with the thicknesses set to zero. The leapfrog row is a one-period transform, after the transient, against the FDFD phasor at 48 steps per period.

Check	Bar	What is compared
CFS profile	10^{-12} relative	σ, κ, α on the node and midpoint lines
Stretched curls, both polarizations	10^{-12} relative	Sparse pair versus the array curls
Zero-thickness PML step	exact match	Unstretched leapfrog, both polarizations
Line current, coarse band	$4 \cdot 10^{-2}$ relative	Interior E_z versus $H_0^{(2)}$
Line current, refined band	2.5× smaller	Same physical band, half the spacing
Bare PEC window	above 0.5 relative	Those interior samples, with no stretch
Leapfrog phasor, both polarizations	$1.5 \cdot 10^{-2}$ relative	48 steps per period versus FDFD

Embedded conductors are checked on the reduced curl. A PEC band that fills $y \geq b_{\text{short}}$ on a taller grid leaves the spectrum of a PEC cavity of height b_{short}. A PMC band whose lower edge sits on an H_x row, $y = (J + \frac{1}{2})\Delta y$, leaves the TMz quarter-wave $k_y = (n + \frac{1}{2})\pi / y_{\text{face}}$. Decoupled electric samples above that cut are zero eigenvalues and are dropped before the comparison. The open rows use a 64×64 grid with $\Delta = 1$, $a = 8$, eight PML cells, and $ka = 1$. The series takes $k = \omega\sqrt{\mu_0 \varepsilon_0}$. The PEC window is the unstretched samples with $\rho \geq a + 2\Delta$. The dielectric rows use that grid with $n = 1.5$. Their window is the unstretched samples with $|\rho - a| \geq 2\Delta$, inside and outside the disk, and the drive is the contrast current of Sec. 4.4.

Check	Bar	What is compared
PEC band, both polarizations	10^{-10} relative	Shortened cavity versus (3.2)
PMC quarter-wave, TMz	10^{-10} relative	Lowest positive eigenvalue, $k_y = \pi / (2 y_{\text{face}})$
Open PEC cylinder, TMz	$8 \cdot 10^{-2}$ relative	E_z, $\rho \geq a + 2\Delta$
Open PEC cylinder, TEz	10^{-1} relative	E_x and E_y, same window
Open dielectric cylinder, TMz	$8 \cdot 10^{-3}$ relative	E_z, $\lvert\rho - a\rvert \geq 2\Delta$
Open dielectric cylinder, TEz	$1.3 \cdot 10^{-2}$ relative	E_x and E_y, same window
Resistive sheet, TMz	$6 \cdot 10^{-4}$ relative	E_z, 8×80, $\Delta = 1$, $Z_s = \eta_0$, $\lambda = 46$
Resistive sheet, TEz	$1.2 \cdot 10^{-2}$ relative	E_x and E_y, same grid
Symmetric sheet, TMz	$4.2 \cdot 10^{-2}$ relative	E_z, same box, face $y = 40.5$, $\chi_{ee} = 0.5$, $\chi_{mm} = 0.25$
Symmetric sheet, TEz	$1.9 \cdot 10^{-1}$ relative	E_x and E_y, same grid

The resistive-sheet rows are a closed PEC box with zero impressed current and the analytic trace on the wall. The sheet is the conductance (4.5) on one Ampere cell, at $y = 40$. The comparison is the packed free electric vector. The analytic coefficients at $Z_s = 100\,\Omega$ and $k = 2\pi/0.3$ match one stored MaxwellMOM value of that sheet. The symmetric-sheet rows use the same box with the sheet on the magnetic face $y = 40.5$, $\chi_{ee} = 0.5$, and $\chi_{mm} = 0.25$. The comparison is again the packed free electric vector. A cell of $\Delta = 1/3$ on that same sheet drops both errors by more than half. The coefficients at $k = 2\pi/0.3$ for this χ match one stored MaxwellMOM value.

The 3D curls use the partial derivatives of Sec. 2.2 on every axis. The cavity row compares the Rayleigh quotient of a known mode with (3.2) after the term $(2\sin(k_z\Delta z/2)/\Delta z)^2$ is added, with $k_z = p\pi/c$. The box is $8 \times 6 \times 5$ cells at $\Delta = 1$, and the modes are $(1, 1, 0)$, $(1, 0, 1)$, and $(0, 1, 1)$. Each has one index equal to zero and a single Cartesian component. The manufactured row recovers a random electric vector on a $6 \times 5 \times 4$ box with spacings 0.5, 0.4, and 0.3.

Check	Bar	What is compared
3D linear curls	10^{-12} absolute	Centered differences of a linear field
3D $\mathrm{curl}_H = \mathrm{curl}_E^{\mathsf{T}}$	10^{-12} absolute	PEC and periodic boxes
3D matrix versus array curl	10^{-12} relative	Same boxes, PEC boundaries held at zero
3D PEC cavity, three pure modes	10^{-10} relative	$8 \times 6 \times 5$ box, versus (3.2) plus z
3D manufactured FDFD solve	10^{-10} relative	Recovered e on a $6 \times 5 \times 4$ box
3D stretched curls	10^{-12} relative	Matrix versus array curl, $6 \times 5 \times 4$ box
3D Mie surface	10^{-8} absolute	Tangential E, H and normal D, B at $\rho = a$
Open dielectric sphere	$1.8 \cdot 10^{-2}$ relative	E, $\lvert\rho - a\rvert \geq 2\Delta$, 18^3 box, PML of 3 cells

The stretched-curl row multiplies each partial derivative by $1/s_w$ and compares the sparse matrix with the array curl. The surface row evaluates the sphere series from both sides of $\rho = a$. Tangential E and H match, as do the normal components of D and B. That series is the outgoing spherical wave $h_n^{(2)}$. Its coefficients are not written out in this note. The open sphere is a contrast-current solve: the staircase ball stays in the permittivity, the unknown is the scattered field, and the comparison keeps the electric samples outside the PML and at least 2Δ off the surface.

5.1 Normal-incidence Fresnel slab

A slab $y_1 \leq y \leq y_2$ of thickness d, relative permittivity ε_r, and $\mu = \mu_0$ sits in vacuum. The incident wave travels toward $+y$ and equals 1 at the front face. With $n = \sqrt{\varepsilon_r}$, $k = \omega/c$, and $\delta = nkd$, a single-interface reflection r produces

$$R = \frac{r(1 - e^{-2j\delta})}{1 - r^2 e^{-2j\delta}}, \qquad \tau = \frac{(1 - r^2)\, e^{-j\delta}}{1 - r^2 e^{-2j\delta}}. \tag{5.1}$$

For TMz the unknown is E_z and $r = (1 - n)/(1 + n)$. The field is

$$E_z(y) = \begin{cases} e^{-jk(y-y_1)} + R\, e^{jk(y-y_1)}, & y \leq y_1, \\ A\, e^{-jnk(y-y_1)} + C\, e^{jnk(y-y_1)}, & y_1 \leq y \leq y_2, \\ \tau\, e^{-jk(y-y_2)}, & y \geq y_2, \end{cases} \tag{5.2}$$

with $A = \frac{1}{2}\big(1 + R + (1 - R)/n\big)$ and $C = \frac{1}{2}\big(1 + R - (1 - R)/n\big)$. Then $H_x = -\partial_y E_z/(j\omega\mu_0)$: a forward vacuum wave has $H_x = E_z/\eta$ and a backward wave has $H_x = -E_z/\eta$, where $\eta = \sqrt{\mu_0/\varepsilon_0}$. Inside, the forward part of H_x carries an extra factor of n.

TEz is a different scalar. Tangential H reflects with $r_H = (n-1)/(n+1) = -r$, and (5.1) is evaluated at r_H. Transmission depends on r^2, so τ is unchanged and $R_H = -R$. Interior amplitudes satisfy $A_H + C_H = 1 + R_H$ and $A_H - C_H = n(1 - R_H)$,

$$H_z(y) = \begin{cases} e^{-jk(y-y_1)} + R_H\, e^{jk(y-y_1)}, & y \le y_1, \\ A_H\, e^{-jnk(y-y_1)} + C_H\, e^{jnk(y-y_1)}, & y_1 \le y \le y_2, \\ \tau\, e^{-jk(y-y_2)}, & y \ge y_2, \end{cases} \tag{5.3}$$

with $A_H = \frac{1}{2}\big(1 + R_H + n(1 - R_H)\big)$ and $C_H = \frac{1}{2}\big(1 + R_H - n(1 - R_H)\big)$. The lift uses $E_x = \partial_y H_z/(j\omega\varepsilon)$ and $E_y = 0$. Outside, a forward wave has $E_x = -\eta H_z$; inside, that factor is η/n. When $\varepsilon_r = 1$, both reflections vanish and $\tau = e^{-jkd}$, so each field equals the incident wave. Continuity of tangential E and H at both faces is the check on these closed forms.

5.2 Dielectric cylinder

The outgoing cylindrical wave is $H_n^{(2)}$, the same convention as MaxwellMOM's two-dimensional PEC cylinder. The incident field e^{-jkx} has the Jacobi–Anger expansion

$$e^{-jkx} = \sum_{n=-\infty}^{\infty} (-j)^n J_n(k\rho)\, e^{jn\varphi}. \tag{5.4}$$

Our TMz unknown is E_z. For a non-magnetic cylinder of radius a and index $m = \sqrt{\varepsilon_r}$, with $\alpha = ka$ and $\beta = mka$,

$$a_n = (-j)^n \frac{m J_n'(\beta)\, J_n(\alpha) - J_n(\beta)\, J_n'(\alpha)}{J_n(\beta)\, H_n^{(2)\prime}(\alpha) - m J_n'(\beta)\, H_n^{(2)}(\alpha)}. \tag{5.5}$$

The interior coefficient follows from continuity of E_z at $\rho = a$. Outside, the scattered field is $a_n H_n^{(2)}(k\rho)\, e^{jn\varphi}$. Inside, it is $b_n J_n(mk\rho)\, e^{jn\varphi}$.

TEz is the H_z problem. Tangential E brings $1/\varepsilon$, so the contrast multiplies the undifferentiated interior Bessel function,

$$a_n = (-j)^n \frac{m J_n(\beta)\, J_n'(\alpha) - J_n'(\beta)\, J_n(\alpha)}{J_n'(\beta)\, H_n^{(2)}(\alpha) - m J_n(\beta)\, H_n^{(2)\prime}(\alpha)}. \tag{5.6}$$

The PEC reduction used as a regression is the separate coefficient $a_n = -(-j)^n J_n(\alpha)/H_n^{(2)}(\alpha)$, not the limit $\varepsilon_r \to \infty$ of (5.5). The TEz form, from $E_\varphi = 0$, is $a_n = -(-j)^n J_n'(\alpha)/H_n^{(2)\prime}(\alpha)$.

The open PEC cylinder is the scattered-field pair in the conductor table above. The open dielectric cylinder is the contrast-current pair in that same table. The open dielectric sphere is the contrast-current row in the 3D table. The resistive sheet and the symmetric sheet are the last rows in the table above the 3D curls. The Bloch cell is in the code: the forward x wrap carries $e^{-jk_{B,x}a}$, and the unit cell is compared with the analytic oblique sheet. A piecewise sheet assigns each cut face its own susceptibilities and is compared with that piece's analytic field. Polynomial electric dispersion, $\chi_{ee0} + \chi_{ee2}k_t^2$, is the normal-susceptibility operator and is compared with the analytic oblique sheet. Polynomial magnetic dispersion, $\chi_{mm0} + \chi_{mm2}k_t^2$, is the second difference of tangential H and is compared the same way. A denominator in k_t remains later.

Prescribed external maps are checked by confirming that the samples the stepper reads are the samples the law was given. They are not a coupled thermal or mechanical solve.

6 GSTC metasurfaces

6.1 Jump conditions

A zero-thickness sheet of normal $\hat{\mathbf{n}}$, pointing from the $-$ side to the $+$ side, replaces the tangential continuity conditions by

$$\hat{\mathbf{n}} \times (\mathbf{H}^+ - \mathbf{H}^-) = \partial_t \mathbf{P}_s, \tag{6.1}$$

$$(\mathbf{E}^+ - \mathbf{E}^-) \times \hat{\mathbf{n}} = \mu_0 \, \partial_t \mathbf{M}_s. \tag{6.2}$$

For a monoanisotropic sheet the surface polarizations are

$$\mathbf{P}_s = \varepsilon_0 \, \chi_{ee} \cdot \mathbf{E}_{\mathrm{av}}, \qquad \mathbf{M}_s = \chi_{mm} \cdot \mathbf{H}_{\mathrm{av}}, \tag{6.3}$$

with the average of the two one-sided tangential fields. In the frequency domain ∂_t is $j\omega$. These are the same jumps as in the symmetric Yee-cell scheme of Smy, Stewart, and Gupta (arXiv:1706.10136) and the same $e^{j\omega t}$ normalization as MaxwellMOM's `rt_from_chi`: there $J = j\omega\varepsilon_0\chi_{ee}E_{\mathrm{av}}$ and $M = j\omega\mu_0\chi_{mm}H_{\mathrm{av}}$.

For TMz and a sheet of normal $\hat{\mathbf{y}}$, (6.1) and (6.2) become

$$-(H_x^+ - H_x^-) = \partial_t P_z, \tag{6.4}$$

$$E_z^+ - E_z^- = -\mu_0 \, \partial_t M_x. \tag{6.5}$$

H_y is the normal field. It is not split.

Bianisotropic coupling extends (6.3) to

$$\mathbf{P}_s = \varepsilon_0 \chi_{ee} \cdot \mathbf{E}_{\mathrm{av}} + c_0^{-1} \chi_{em} \cdot \mathbf{H}_{\mathrm{av}}, \qquad \mathbf{M}_s = \chi_{mm} \cdot \mathbf{H}_{\mathrm{av}} + \eta_0^{-1} \chi_{me} \cdot \mathbf{E}_{\mathrm{av}}. \tag{6.6}$$

A reciprocal sheet uses $\chi_{me} = \chi_{em}$. That equality is the scalar form of the jump match in MaxwellMOM's `rt_from_chi_omega`. The reflective lock is $\chi_{em} = \chi_{me} = -2j/k$ with the other susceptibilities zero, and at normal incidence it gives $R = 1$, $T = 0$. On both polarizations the frequency-domain magnetic jump gains a term $-(j\omega/c)\chi_{em}H_{\mathrm{av}}$ and the electric jump gains $(j\omega/c)\chi_{me}E_{\mathrm{av}}$. χ_{mm}^{nn} stays on the TMz magnetic jump. When $\chi_{em} = \chi_{me} = 0$ the coefficients remain (6.9) and (6.10).

6.2 Reflection and transmission

Take a uniform sheet at $y = 0$, host medium (ε, μ) on both sides, and a TMz wave at angle θ from the normal. The reflected and transmitted amplitudes are defined by $E_z^+ = TE_0$ and $E_z^- = E_0(1 + R)$ at the sheet, with the incident wave coming from $y < 0$. Substituting the plane waves into (6.4) and (6.5), with constant susceptibilities, gives

$$R + T = \frac{1 - \tilde{\alpha}}{1 + \tilde{\alpha}}, \qquad T - R = \frac{1 - \tilde{\beta}}{1 + \tilde{\beta}}, \tag{6.7}$$

where $\alpha_0 = jk\chi_{ee}/2$, $\beta_0 = jk\chi_{mm}/2$, $k = \omega/c$, and

$$\tilde{\alpha} = \alpha_0 / \cos\theta, \qquad \tilde{\beta} = \beta_0 \cos\theta. \tag{6.8}$$

Hence

$$R = \frac{1}{2}\left(\frac{1 - \tilde{\alpha}}{1 + \tilde{\alpha}} - \frac{1 - \tilde{\beta}}{1 + \tilde{\beta}}\right), \tag{6.9}$$

$$T = \frac{1}{2}\left(\frac{1 - \tilde{\alpha}}{1 + \tilde{\alpha}} + \frac{1 - \tilde{\beta}}{1 + \tilde{\beta}}\right). \tag{6.10}$$

At $\theta = 0$ the tildes drop off and (6.9)–(6.10) are MaxwellMOM's normal-incidence formula. TMz with the sheet normal $\hat{\mathbf{y}}$ and the plane of incidence the xy-plane is s polarization, which that code calls TE.

TEz on the same sheet is p polarization (TM in MaxwellMOM). The cosines trade places,

$$\tilde{\alpha} = \alpha_0 \cos\theta, \qquad \tilde{\beta} = \beta_0 / \cos\theta, \tag{6.11}$$

and R, T keep the form (6.9)–(6.10). A normal magnetic susceptibility χ^{nn}_{mm} enters only the s/TE electric coefficient,

$$\tilde{\alpha} \leftarrow \tilde{\alpha} + \frac{jk \sin^2\theta \, \chi^{nn}_{mm}}{2\cos\theta}, \tag{6.12}$$

which is the correction MaxwellMOM applies for a flat sheet. The coded TMz jump keeps H_y unsplit and adds $\chi^{nn}_{mm} \partial_x H_{y,\mathrm{av}}$ to (6.13). Equation (6.12) is the plane-wave reduction of that term. TEz does not carry it. Nonzero χ_{em} and χ_{me} replace the split (6.7) by the same two jump rows. Their coefficients are $jk\chi_{em}/2$ and $jk\chi_{me}/2$, with no extra factor of $\cos\theta$.

6.3 Symmetric cell

The sheet sits halfway between electric nodes E_z^- at index j and E_z^+ at index $j+1$. The magnetic sample that would have occupied that face is replaced by a pair H_x^-, H_x^+ on the two sides of a zero-thickness cut. The averages and jumps in (6.4) and (6.5) use this pair and the two electric nodes. Bulk Faraday updates for H_y, and for the H_x faces that are not the cut, are unchanged: their stencils do not cross the sheet.

In the frequency domain the cut and the two Ampere cells that touch it are the local system

$$H_x^+ - H_x^- + j\omega\varepsilon_0\chi_{ee}\frac{E_z^- + E_z^+}{2} = 0, \tag{6.13}$$

$$E_z^+ - E_z^- + j\omega\mu_0\chi_{mm}\frac{H_x^- + H_x^+}{2} = 0, \tag{6.14}$$

$$\frac{H_y|_{i,j} - H_y|_{i-1,j}}{\Delta x} - \frac{H_x^- - H_x^{\mathrm{below}}}{\Delta y} - j\omega\varepsilon E_z^- = J_z^-, \tag{6.15}$$

$$\frac{H_y|_{i,j+1} - H_y|_{i-1,j+1}}{\Delta x} - \frac{H_x^{\mathrm{above}} - H_x^+}{\Delta y} - j\omega\varepsilon E_z^+ = J_z^+. \tag{6.16}$$

H_x^{below} and H_x^{above} are the ordinary Yee samples one cell off the cut. They are known from the bulk Faraday update when the cell is marched in time, and they are ordinary unknowns when the same rows are assembled into (4.1). Equations (6.13) and (6.14) replace the Faraday row that used to connect E_z^- to E_z^+ across one Δy.

For a non-dispersive real χ the time-domain statement is the same pair with $j\omega$ replaced by ∂_t acting on $\mathbf{P}_s$ and $\mathbf{M}_s$. The four surface unknowns at a given step are coupled, so the cell is a 4×4 (or a 2×2 after eliminating $H_x^{\pm}$) solve, while the rest of the grid stays an explicit leapfrog. That local solve is the one described for the symmetric cell in arXiv:1706.10136.

Two other placements are on record there. The asymmetric cell inserts the sheet between an existing H sample and an existing E sample and adds no nodes. The tight asymmetric cell adds one electric node on one side of the cut and one magnetic node on the other, and it is the placement that paper found to converge fastest. Milestone 5 codes (6.13)–(6.16) and checks R and T against (6.9) and (6.10). On the validation grid the half-cell offset is the leading error, and refining the cell drops it, so this placement stays. The tight asymmetric cell remains the recorded alternative. The coded cell appends $H^{\pm}$ after the electric unknowns. It does not paint the conductance (4.5).

A constant real χ, independent of ω, is a convenient FDFD value and a legal reduction of (6.13)–(6.14). It is not a causal broadband model. Broadband FDTD uses a Lorentz oscillator

for each resonance,

$$\chi_{ee}(\omega) = \frac{\omega_{ep}^2}{\omega_{e0}^2 - \omega^2 + j\alpha_e\omega}, \qquad \partial_t^2 P_z + \alpha_e \partial_t P_z + \omega_{e0}^2 P_z = \varepsilon_0 \omega_{ep}^2 E_{z,\mathrm{av}}, \tag{6.17}$$

$$\chi_{mm}(\omega) = \frac{\omega_{mp}^2}{\omega_{m0}^2 - \omega^2 + j\alpha_m\omega}, \qquad \partial_t^2 M_x + \alpha_m \partial_t M_x + \omega_{m0}^2 M_x = \omega_{mp}^2 H_{x,\mathrm{av}}. \tag{6.18}$$

Several poles superpose. The auxiliary variables are marched with the local cell solve. This is the dispersive extension, after the constant-χ sheet matches (6.9) and (6.10).

6.4 Piecewise $\chi(x)$ and what remains

The Bloch cell is in the code. Stored samples are the physical field on $[0, a)$. The forward x wrap multiplies by $e^{-jk_{B,x}a}$ and the backward wrap by $e^{+jk_{B,x}a}$. The y wrap stays at phase one. The same factors enter the wrapped neighbors of $\chi_{mm}^{nn}\partial_x H_y$ and the wrapped TEz link at the right edge. A uniform sheet on that cell is specular: the orders $m = \pm 1$ stay below 10^{-8} of the specular amplitude, beside the analytic reflection and transmission. A piecewise sheet is in the code. Each cut sample takes the susceptibilities of the piece that contains its x, including χ_{mm2} when that piece sets it, and the jumps stay the same. A gap between pieces leaves those magnetic samples in the bulk curl. Polynomial electric dispersion is in the code. $\chi_{ee}(k_t) = \chi_{ee0} + \chi_{ee2}k_t^2$ is the normal susceptibility $\chi_{mm}^{nn} = \chi_{ee2}k^2$, MaxwellMOM's multi-$k_t$ reduction of that polynomial. TMz carries the term and TEz keeps χ_{ee0}. Polynomial magnetic dispersion is in the code. $\chi_{mm}(k_t) = \chi_{mm0} + \chi_{mm2}k_t^2$ replaces $\chi_{mm}H$ on the electric jump by $\chi_{mm0}H - \chi_{mm2}\partial_{xx}H$, in both polarizations. The discrete form is the centered second difference of the appended $H^{\pm}$, with the same Bloch weights as the normal term. A neighbor that leaves the cut or meets the PEC wall is refused when $\chi_{mm2} \neq 0$. The analytic target uses the continuous k_t. A denominator in k_t and Lorentz poles on the sheet remain later.

7 Software architecture

7.1 Modules

The modules are the layout the tests exercise.

Module	Responsibility
`grid.yee2d`	Counts, spacings, polarization, boundary, shapes, coordinates
`materials.volume`	Uniform, staircase, and mapped ε, μ, σ
`materials.dispersion`	Debye and Lorentz laws, auxiliaries J_p
`materials.conductors`	Staircase PEC and PMC masks
`operators.curl2d`	`curl_e`, `curl_h`, array curls, DOF order, optional $1/s_w$
`operators.pml`	Stretch (3.7)–(3.9); 2D auxiliary and 3D FDFD
`drivers.fdtd2d`	Leapfrog step, J and J_p, invariant (3.6)
`drivers.fdfd2d`	System (4.1), Dirichlet lift, TF/SF line, optional sheet
`analytics.cavity2d`	Frequency (3.2) and mode shapes (3.4), (3.5)
`analytics.dispersion`	Frequency (3.3) and the three-sample recurrence
`analytics.fresnel`	Slab fields (5.2), (5.3)
`analytics.mie2d`	Cylinder series (5.5), (5.6)
`analytics.green2d`	Outgoing line current, $H_0^{(2)}$ and $H_1^{(2)}$
`analytics.beam2d`	2D paraxial Gaussian beam
`analytics.farfield2d`	2D TMz near-to-far integral
`analytics.dipole3d`	Hertzian dipole, outgoing spherical wave
`grid.yee3d`	3D counts, spacings, boundary, shapes, coordinates
`materials.volume3d`	Uniform law and a staircase ball
`operators.curl3d`	3D curls and array curls, optional $1/s_w$
`drivers.fdfd3d`	System (4.1), optional PML
`analytics.cavity3d`	Frequency (3.2) plus the z term
`analytics.mie3d`	Sphere series with $h_n^{(2)}$
`materials.sheets`	Resistive σ_{eff}, tangential, normal, magneto-electric, and piecewise χ
`operators.gstc`	Symmetric cell, append $H^{\pm}$
`analytics.sheets`	R, T from (6.9), (6.10), and the magneto-electric jumps

`YeeGrid2D` owns n_x, n_y, Δx, Δy, the polarization, and the boundary kind. Both drivers read shapes from it. A validation case is run in time or in frequency by constructing the other driver on the same object. `YeeGrid3D` owns the three counts, the three spacings, and the same boundary kind. The 3D grid is its own object: every component that the stencil staggers in z carries that stagger. Its FDFD driver reads the same elimination, (4.1), on (E_x, E_y, E_z). An optional PML rebuilds those curls with the same $1/s_w$ as (3.7). A staircase ball assigns one isotropic permittivity to every sample inside the closed sphere.

`UniformIsotropic.sample_tm` and `sample_te` take the grid and optional `omega` and `state` arguments. Milestone 1 passes neither. A dispersive or externally modulated law implements the same methods and returns arrays of the same shapes. The stepper does not learn about temperature.

A resistive sheet is not a new unknown. `ResistiveSheet` adds $1/(Z_s \Delta y)$ to σ on the tangential electric samples of the Ampere cell in Sec. 4.3. Both drivers then use that σ unchanged. The symmetric cell of Sec. 6.3 replaces the magnetic sample on the cut with the pair H^-, H^+. `operators.gstc` appends that pair, and the solve returns the electric prefix. PEC remains a boundary kind rather than a material, because it removes unknowns instead of filling coefficients. An embedded PEC or PMC applies that removal to interior samples, through `materials.conductors`.

7.2 Relation to MaxwellMOM

MaxwellMOM is a surface integral code: RWG bases, Green's functions, and dense blocks. MaxwellFD does not import it. The shared pieces are the convention and the analytic targets.

- $e^{j\omega t}$, and the four SI constants, copied numerically.

- Sheet reflection and transmission, (6.9) and (6.10), including the oblique factors and the χ_{mm}^{nn} correction, which are the formulas behind `rt_from_chi`. The magneto-electric pair is checked against one stored value of `rt_from_chi_omega`.
- The map (4.4) from a surface impedance to χ_{ee}.
- Mie and Fresnel results as external checks, not as shared functions.

A number computed by MaxwellMOM for a configuration both codes can run (a flat sheet, a periodic cell) is a regression oracle. It is stored as a literal in a test, not read from the other repository at runtime.

7.3 Milestone map

1. **This slice.** 2D grid, both drivers, PEC cavities, periodic dispersion, invariant (3.6).
2. Volumetric $\varepsilon(\mathbf{r})$, then $\varepsilon(\omega)$: complex permittivity in FDFD, Debye or Lorentz auxiliaries in FDTD. Slab versus Fresnel, cylinder versus 2D Mie. External maps enter only as arrays the law samples.
3. PML from Sec. 3.5, embedded PEC and PMC objects, an open PEC cylinder, and an open dielectric cylinder, both against the Mie series. The 3D cell, a rectangular PEC cavity, and an open dielectric sphere small enough to factor are in the code.
4. Resistive sheet, Sec. 4.3. The conductance (4.5) is in the code. It is compared with (6.9), (6.10), and one stored MaxwellMOM value.
5. Symmetric GSTC cell, Sec. 6.3. Tangential χ_{ee} and χ_{mm}, the normal susceptibility χ_{mm}^{nn}, and the magneto-electric pair χ_{em}, χ_{me} are in the code and are compared with the analytic reflection and transmission. A piecewise run assigns each cut sample the five susceptibilities of the piece that contains its x. The jumps stay the same. Polynomial electric dispersion $\chi_{ee0} + \chi_{ee2}k_t^2$ is the same normal susceptibility with $\chi_{mm}^{nn} = \chi_{ee2}k^2$. Polynomial magnetic dispersion $\chi_{mm0} + \chi_{mm2}k_t^2$ is a second difference of the appended tangential H, in both polarizations.
6. Hertzian dipole, compared with the outgoing spherical wave, a 2D paraxial Gaussian beam on a total-field/scattered-field split, and a 2D TMz near-to-far integral compared with the leading Hankel wave of a line current. A Bloch phase $e^{-jk_{B,x}a}$ on the forward periodic x wrap is in the code. A unit cell narrower than one transverse period is compared with the analytic oblique sheet.
7. Profiling, Numba, iterative FDFD, GPU. A denominator in $\chi(k_t)$ and Lorentz poles on the sheet remain later.

Acceleration does not change the curls. Numba and CuPy, when they arrive, replace the array update and the sparse matrix-vector product. The direct factorization remains the 2D reference.

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